

Gene flux erodes the karyotype barrier

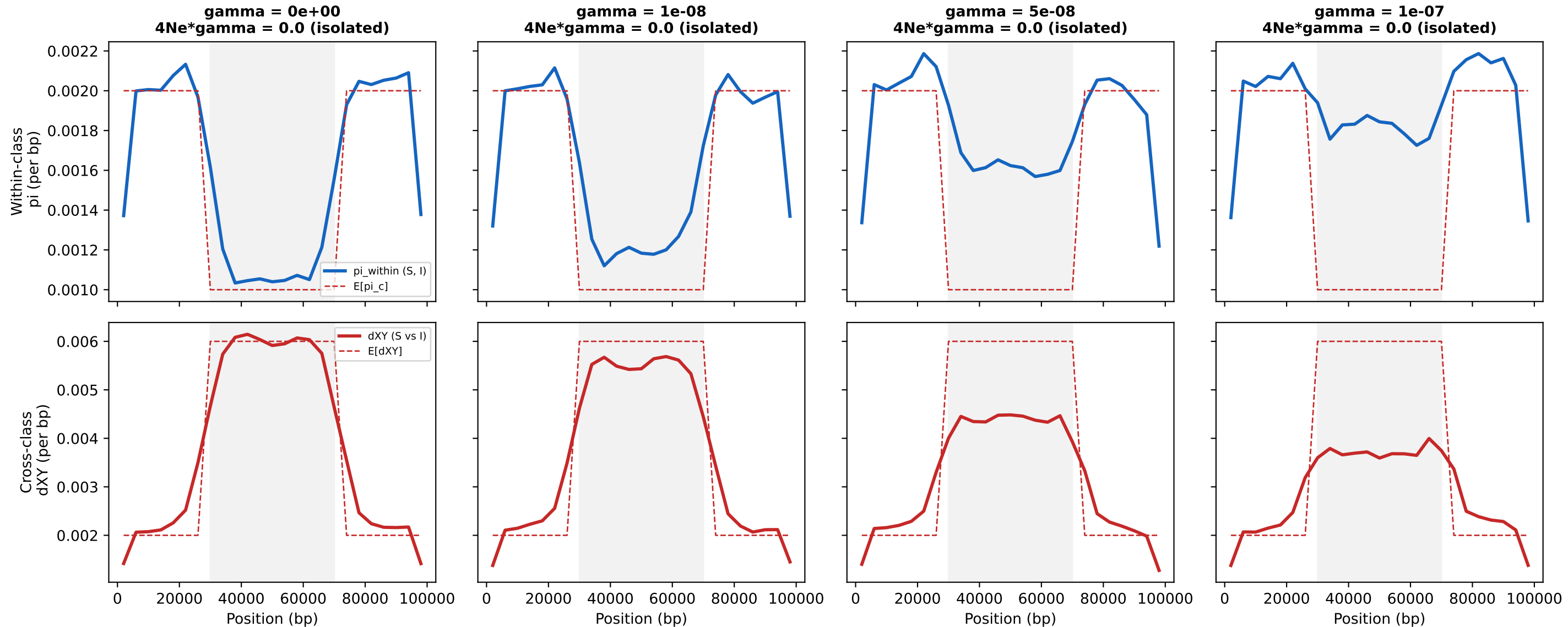


Figure. Effect of gene-conversion rate γ on the karyotype barrier. Each column shows a different γ (0, $1e-8$, $5e-8$, $1e-7$ bp-1 gen-1). (Top) Average within-class diversity π . Dashed: structured coalescent prediction ($p*\theta$ inside inv, θ outside). (Bottom) Cross-class dXY . Dashed: $2*\mu*(t_{inv}+2N_e)$ inside, $2*\mu*2N_e$ outside. As γ increases, gene flux erodes the barrier — π recovers and dXY collapses, first at the centre (where $\phi(x)$ peaks) and last at breakpoints. Parameters: $N_e=50,000$, $p_{inv}=0.5$, $t_{inv}=200,000$ gen, $L=100$ kb, $\mu=1e-8$, $r=1e-8$, 40 reps. Command: `pixi run -e all python examples/gene_flux_demo.py`